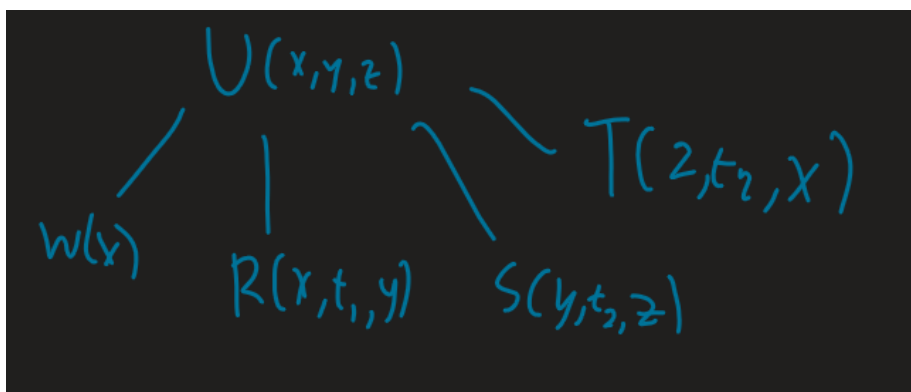


hw2

Ian Chi

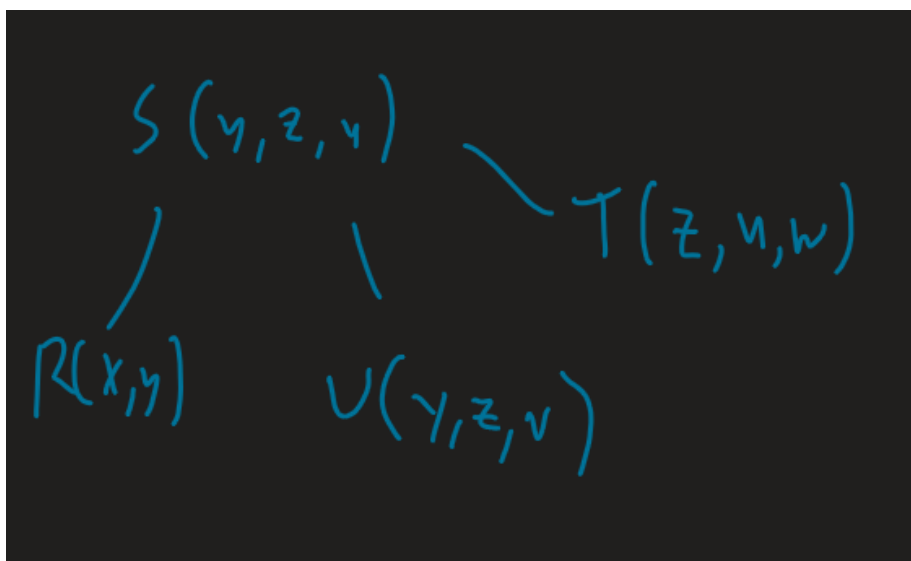
February 6, 2026

1



1.

2. S, R, T form a cycle. S, R contain z . R, T contain u . T, S contain x , this forms a cycle.



3.

2

```
1.      SELECT COUNT(*)
        FROM edges e1
        JOIN edges e2 ON e1.dst = e2.src
        JOIN edges e3 ON e2.dst = e3.src
        JOIN edges e4 ON e3.dst = e4.src
        JOIN edges e5 ON e4.dst = e5.src
        WHERE e1.src % 100 = 0
           AND e5.dst % 100 = 0;
```

it took 3:19 for the full graph

2. $E(x,y) - E(y,z) - E(z,w) - E(w,u) - E(u,v)$

```
3.      WITH
        E1 AS (SELECT * FROM edges WHERE src % 100 = 0),
        E5 AS (SELECT * FROM edges WHERE dst % 100 = 0),

        E4b AS (SELECT * FROM edges WHERE dst IN (SELECT src FROM E5)),
        E3b AS (SELECT * FROM edges WHERE dst IN (SELECT src FROM E4b)),
        E2b AS (SELECT * FROM edges WHERE dst IN (SELECT src FROM E3b)),
        E1b AS (SELECT * FROM E1 WHERE dst IN (SELECT src FROM E2b)),

        E2t AS (SELECT * FROM E2b WHERE src IN (SELECT dst FROM E1b)),
        E3t AS (SELECT * FROM E3b WHERE src IN (SELECT dst FROM E2t)),
        E4t AS (SELECT * FROM E4b WHERE src IN (SELECT dst FROM E3t)),
        E5t AS (SELECT * FROM E5 WHERE src IN (SELECT dst FROM E4t))

        SELECT COUNT(*)
        FROM E1b e1
        JOIN E2t e2 ON e1.dst = e2.src
        JOIN E3t e3 ON e2.dst = e3.src
        JOIN E4t e4 ON e3.dst = e4.src
        JOIN E5t e5 ON e4.dst = e5.src;
```

4. Time: 177973.167 ms (02:57.973) the time maybe is saved a tiny bit but
pgsql probably already does some optimization

3

```
1.      for each z:
           count_z[z] = number of tuples (z, u) in E3
```

```

for each y:
    count_y[y] = 0
    for each (y, z) in E2 with this y:
        count_y[y] += count_z[z]

total = 0
for each (x, y) in E1:
    total += count_y[y]

output total

```

```

2. WITH
   z_count AS (
       SELECT src AS z, COUNT(*) AS cz
       FROM edges
       GROUP BY src
   ),

   y_count AS (
       SELECT e.src AS y, SUM(zc.cz) AS cy
       FROM edges e
       JOIN z_count zc ON e.dst = zc.z
       GROUP BY e.src
   )

   SELECT SUM(yc.cy) AS path_count
   FROM edges e
   JOIN y_count yc ON e.dst = yc.y;

```

3. Time: 316.340 ms vs Time: 177832.565 ms (02:57.833)